

Math 289, Fall 2026

Homework 4

Due Wednesday, September 30, 2026, 3:00pm, on Gradescope

Problem 1. Find the integers n for which

$$\frac{n^3 - 3n^2 + 4}{2n - 1}$$

is an integer.

Problem 2. Solve in positive integers the equation

$$x^{x+y} = y^{y-x}.$$